

# Research Statement

Guangxu Yang

My research lies in theoretical computer science, with a focus on computational complexity theory for its combination of mathematical depth, extensive and surprising connections to other disciplines, and broad relevance to real-world computing. My work has focused on communication complexity with a particular emphasis on developing new techniques for proving communication lower bounds and understanding the quantum advantages in communication and bounded-memory settings.

My doctoral research has developed along three closely related directions:

- I developed a gadgetless lifting framework that extends query-to-communication lifting to problems without an explicit gadget. Together with my coauthors, I used this framework to resolve several open problems in proof complexity [YZ24], communication complexity [MYZ25], cryptography [HMW<sup>+</sup>25, HMYZ23].
- I proposed a lifting framework connecting two-party communication with the Number-on-Forehead (NOF) model. In joint work, we used this framework to obtain the first exponential separation between quantum and randomized one-way NOF communication [YZ26] answering the open problems proposed by Gavinsky and Pudlák [GP08], and achieved optimal separations between deterministic and randomized one-way NOF communication [YZ25].
- My coauthors and I proved an exponential quantum space advantage for approximating Max- $k$ SAT in the streaming setting [WY26]; combined with the known results, it gives a complete classification of quantum space advantages for all Boolean Max-2CSPs in the streaming setting. We also developed a matrix-discrepancy approach to prove quantum streaming lower bounds [GLYZ26].

Below, I describe some of my current work in detail, followed by future plans.

## Gadgetless Lifting for Number-in-Hand Communication

Communication complexity studies how much communication is required for several parties to compute a function of distributed inputs. In the standard two-party model, Alice receives  $x$ , Bob receives  $y$ , and they communicate to compute  $f(x, y)$ . The model can be viewed as a generalization of information theory and has wide-ranging applications in areas such as circuit complexity, proof complexity, streaming algorithms, data structures, and cryptography.

Query-to-communication lifting theorems are among the most powerful techniques in communication complexity [GP18, GLM<sup>+</sup>16, GPW18, GPW20]. A lifting theorem starts with a query problem  $f : \{0, 1\}^n \rightarrow \{0, 1\}$  and replaces each input bit with a small two-party gadget  $g : \{0, 1\}^b \times \{0, 1\}^b \rightarrow \{0, 1\}$ , producing the communication problem  $f \circ g^n$ . It then characterizes the communication complexity of  $f \circ g^n$  up to constant factors in terms of the query complexity of  $f$ . Conceptually, such a theorem converts an arbitrary communication protocol for  $f \circ g^n$  into a structured protocol that behaves like a decision tree for  $f$ . Despite their broad impact, existing lifting theorems have two important limitations. First, they require large gadgets, typically with  $b = \Omega(\log n)$ ; obtaining query-to-communication lifting theorems for constant-size gadgets remains a major open problem. Second, many natural communication problems do not have the composed form  $f \circ g^n$ .

## My Work

To move beyond explicit gadgets, my collaborators and I introduced a gadgetless lifting framework. Instead of embedding a problem into  $f \circ g^n$ , we identify a problem-specific class of structured protocols and show that every communication protocol can be simulated by one in this class. In several applications, the simulation relies on a structure-versus-pseudorandomness decomposition inspired by [GPW20].

This framework has yielded several concrete results. In joint work with Jiapeng Zhang, I proved communication lower bounds for collision problems, resolving open problems in proof complexity and cryptography [YZ24]. With Xinyu Mao and Jiapeng Zhang, I applied gadgetless lifting to the round-communication tradeoffs and proved improved lower bounds for pointer chasing that go beyond standard round-elimination techniques [MYZ25]. With Mi-Ying Huang, Xinyu Mao, Shuo Wang, and Jiapeng Zhang, I extended the framework to multiparty communication, with applications to streaming lower bounds [HMY<sup>+</sup>25]. Related ideas also yielded lower bounds for key-agreement protocols in cryptography [HMY23].

The broader contribution of this line of work is methodological: it recasts lifting not merely as a theorem about gadget composition, but as a general strategy for uncovering and exploiting structure in communication protocols.

## Future Directions

I plan to extend the gadgetless-lifting perspective to circuit complexity through the Karchmer–Wigderson (KW) game correspondence between bounded-depth circuits and two-party protocols [KW88]. A long-standing challenge in circuit complexity is to develop broadly applicable top-down methods for proving strong lower bounds against bounded-depth circuits [HJP95, MW19, GRSS23]. I will pursue this direction by lifting arbitrary KW protocols to more structured protocols that are easier to analyze, using the classical lower bound  $\text{PARITY} \notin \text{AC}^0$  as an initial test case. The main technical challenge is to identify, in the KW setting, an analogue of the structure-versus-pseudorandomness decomposition underlying gadgetless lifting [GPW20].

## Quantum Advantages in Number-on-Forehead Communication

A fundamental objective in the study of quantum advantage is to exhibit computational separations across diverse models. Communication complexity has seen substantial progress toward this goal: a line of work [BCW98, Raz99, BYJK04, GKK<sup>+</sup>08, Gav16, GGJL25] now establishes exponential separations in several two-party communication settings. Despite these successes in two-party communication complexity, quantum advantage in the multiparty setting remains poorly understood. The premier model for multiparty communication is the Number-on-the-Forehead (NOF) model, introduced by Chandra, Furst, and Lipton [CFL83]. In this model, the input is partitioned into  $k$  parts, and player  $i$  can see all parts except their own (as if they were "written on their forehead"). Recently, Göös, Gur, Jain, and Li [GGJL25] highlighted the following open problem: Is there an exponential quantum advantage in Number-on-Forehead communication?

Resolving this problem would demonstrate that quantum communication retains an exponential advantage in a multiparty setting with highly overlapping input views. The main obstacle to obtaining strong separations between quantum and randomized protocols in the NOF model, already noted in [GGJL25], is the lack of techniques for proving lower bounds for randomized NOF protocols. Most existing methods, such as the generalized discrepancy method [RY15], apply equally

well to quantum and randomized communication, making it difficult to distinguish the power of quantum versus classical protocols in the NOF setting [LS09].

## My Work

As a first step toward understanding quantum advantage in NOF communication, I studied its one-way variant, in which each player speaks exactly once in a fixed order. The one-way NOF model holds pivotal importance for several long-standing open problems in circuit complexity [HG90, BT94, Cha07] and wide applications such as additive combinatorics, cryptography, distributed computing, and streaming algorithms. Gavinsky and Pudlák [GP08], asked for an explicit exponential separation between quantum and randomized classical one-way NOF communication.

In the joint works with Jiapeng Zhang, I developed a new lifting framework that connects two-party communication to one-way NOF communication. Using this framework, we obtained the first exponential separation between quantum and randomized one-way NOF communication [YZ26], resolving an open problem posed by Gavinsky and Pudlák [GP08].

Using this framework, we further established a deterministic lifting theorem that converts two-party one-way communication lower bounds into one-way NOF lower bounds [YZ25] and proved the optimal separation between deterministic and randomized one-way NOF communication for constant number of players.

## Future Directions

Our ultimate goal is to prove an exponential separation between quantum and randomized NOF communication [GGJL25]. Toward this goal, we plan to strengthen our lifting framework for unrestricted NOF protocols; the same framework may also yield an optimal separation between deterministic and randomized NOF communication [KLM24]. A complementary direction is to move beyond the specially constructed search problems underlying our current one-way separations and identify natural problems that exhibit exponential quantum advantages in the one-way NOF model.

## Quantum Space Advantages in Streaming Setting

Quantum algorithms are often motivated by potential speedups, but proving unconditional exponential time separations remains notoriously difficult. Space complexity offers a different and especially natural lens on quantum advantage: quantum memory is physically scarce, and the streaming model permits unconditional lower bounds on classical space complexity. In this model, the input arrives sequentially, and each update or constraint must be processed using memory much smaller than the input size. The streaming model is closely connected to one-way communication complexity, as the algorithms memory state can be viewed as the message passed between players processing consecutive portions of the stream.

Early work leveraged exponential quantum advantages in one-way communication to obtain analogous space advantages for specially constructed streaming problems [GKK<sup>+</sup>08]. More recent work extended this phenomenon to natural optimization problems, showing that Maximum Directed Cut, a natural Boolean Max-2CSP, admits an exponential quantum space advantage [KPV24]. This development raises a broader classification question: Which Max-CSPs admit exponential quantum space advantages in the streaming model?

## My Work

Motivated by this open question, I studied quantum space advantages for Boolean Max-CSPs, starting with Max- $k$ SAT. Max- $k$ SAT is one of the most basic Boolean CSPs, and classical streaming lower bounds of Chou, Golovnev, and Velusamy [CGV20] imply that any algorithm achieving an approximation ratio better than  $\sqrt{2}/2 \approx 0.7071$  requires  $\Omega(\sqrt{n})$  space. Thus, a polylogarithmic-space quantum streaming algorithm with even a modest improvement over this threshold would yield an exponential quantum space advantage.

In joint work with Haoyu Wang, I proved exponential quantum space advantage for Max- $k$ SAT [WY26]. We give a one-pass quantum streaming algorithm using  $\text{polylog}(n)$  space that achieves a 0.7172-approximation on instances with  $n$  variables. Combined with the classical lower bound of [CGV20], this gives an exponential separation between quantum and classical streaming space for Max- $k$ SAT. We further give a one-pass quantum streaming algorithm for Max-2OR using  $\text{polylog}(n)$  space and achieving a 0.7425-approximation. Together with known results, this gives a complete classification of which Boolean Max-2CSPs admit exponential quantum space advantages.

In joint work with In joint work with Minbo Gao, Chenghua Liu, and Tianyi Zhang, I developed a new technique for proving quantum streaming lower bounds via matrix-discrepancy [GLYZ26]. Our approach gives quantum streaming lower bounds for triangle finding. This direction complements the algorithmic side of quantum streaming.

## Future Directions

The results above show that quantum sketching can yield exponential one-pass space advantages for natural Max-CSPs. One particularly important and wide-open direction is to prove multipass quantum streaming lower bounds for Max-CSPs, such as Max-DiCut. Such results would clarify the extent to which quantum space advantages persist beyond the one-pass setting.

Another direction is to bridge asymptotic quantum space advantages and their realization on quantum hardware. Recent work has implemented the quantum pair sketch on a trapped-ion quantum processor and used it to execute a streaming algorithm for Hidden Matching, providing an experimental proof of principle for quantum streaming; it also analyzes the overhead required for fault-tolerant execution [NCK<sup>+</sup>25]. Building on this progress, I plan to develop implementation-oriented variants of our Max- $k$ SAT algorithm and related Max-DiCut algorithms, optimizing logical-qubit count, circuit depth, noise robustness, and classical-control latency.

## References

- [BCW98] Harry Buhrman, Richard Cleve, and Avi Wigderson. Quantum vs. classical communication and computation. In *STOC*, pages 63–68, 1998.
- [BT94] Richard Beigel and Jun Tarui. On ACC. *Computational Complexity*, 4(4):350–366, 1994.
- [BYJK04] Ziv Bar-Yossef, Thathachar S Jayram, and Iordanis Kerenidis. Exponential separation of quantum and classical one-way communication complexity. In *STOC*, pages 128–137, 2004.
- [CFL83] Ashok K Chandra, Merrick L Furst, and Richard J Lipton. Multi-party protocols. In *STOC*, pages 94–99, 1983.
- [CGV20] Chi-Ning Chou, Alexander Golovnev, and Santhoshini Velusamy. Optimal streaming approximations for all boolean Max-2CSPs and Max- $k$ SAT. In *FOCS*, pages 330–341, 2020.
- [Cha07] Amit Chakrabarti. Lower bounds for multi-player pointer jumping. In *CCC*, pages 33–45, 2007.

- [Gav16] Dmitry Gavinsky. Entangled simultaneity versus classical interactivity in communication complexity. In *STOC*, pages 877–884, 2016.
- [GGJL25] Mika Göös, Tom Gur, Siddhartha Jain, and Jiawei Li. Quantum communication advantage in TFNP. In *STOC*, pages 1465–1475, 2025.
- [GKK<sup>+</sup>08] Dmitry Gavinsky, Julia Kempe, Iordanis Kerenidis, Ran Raz, and Ronald de Wolf. Exponential separation for one-way quantum communication complexity, with applications to cryptography. *SIAM Journal on Computing*, 38(5):1695–1708, 2008.
- [GLM<sup>+</sup>16] Mika Göös, Shachar Lovett, Raghu Meka, Thomas Watson, and David Zuckerman. Rectangles are nonnegative juntas. *SIAM Journal on Computing*, 45(5):1835–1869, 2016.
- [GLYZ26] Minbo Gao, Chenghua Liu, Guangxu Yang, and Tianyi Zhang. Quantum communication lower bounds for search problems via matrix discrepancy, 2026. arXiv:2607.08517; ECCC TR26-117.
- [GP08] Dmitry Gavinsky and Pavel Pudlák. Exponential separation of quantum and classical non-interactive multi-party communication complexity. In *CCC*, pages 332–339, 2008.
- [GP18] Mika Göös and Toniann Pitassi. Communication lower bounds via critical block sensitivity. *SIAM Journal on Computing*, 47(5):1778–1806, 2018.
- [GPW18] Mika Göös, Toniann Pitassi, and Thomas Watson. Deterministic communication vs. partition number. *SIAM Journal on Computing*, 47(6):2435–2450, 2018.
- [GPW20] Mika Göös, Toniann Pitassi, and Thomas Watson. Query-to-communication lifting for BPP. *SIAM Journal on Computing*, 49(4):441–461, 2020.
- [GRSS23] Mika Göös, Artur Riazanov, Anastasia Sofronova, and Dmitry Sokolov. Top-down lower bounds for depth-four circuits. In *FOCS*, pages 1048–1055, 2023.
- [HG90] Johan Håstad and Mikael Goldmann. On the power of small-depth threshold circuits. In *FOCS*, pages 610–618, 1990.
- [HJP95] Johan Håstad, Stasys Jukna, and Pavel Pudlák. Top-down lower bounds for depth-three circuits. *Computational Complexity*, 5(2):99–112, 1995.
- [HMW<sup>+</sup>25] Mi-Ying (Miryam) Huang, Xinyu Mao, Shuo Wang, Guangxu Yang, and Jiapeng Zhang. A min-entropy approach to multi-party communication lower bounds. In *CCC*, pages 33:1–33:29, 2025.
- [HMYZ23] Mi-Ying Huang, Xinyu Mao, Guangxu Yang, and Jiapeng Zhang. Communication lower bounds of key-agreement protocols via density increment arguments. In *TCC*, pages 185–206, 2023.
- [KLM24] Zander Kelley, Shachar Lovett, and Raghu Meka. Explicit separations between randomized and deterministic number-on-forehead communication. In *STOC*, pages 1299–1310, 2024.
- [KPV24] John Kallaugher, Ojas Parekh, and Nadezhda Voronova. Exponential quantum space advantage for approximating maximum directed cut in the streaming model. In *STOC*, pages 1805–1815, 2024.
- [KW88] Mauricio Karchmer and Avi Wigderson. Monotone circuits for connectivity require super-logarithmic depth. In *STOC*, pages 539–550, 1988.
- [LS09] Troy Lee and Adi Shraibman. Lower bounds in communication complexity. *Foundations and Trends in Theoretical Computer Science*, 3(4):263–399, 2009.
- [MW19] Or Meir and Avi Wigderson. Prediction from partial information and hindsight, with application to circuit lower bounds. *Computational Complexity*, 28(2):145–183, 2019.
- [MYZ25] Xinyu Mao, Guangxu Yang, and Jiapeng Zhang. Gadgetless lifting beats round elimination: Improved lower bounds for pointer chasing. In *ITCS*, pages 75:1–75:14, 2025.

- [NCK<sup>+</sup>25] Pradeep Niroula, Shouvanik Chakrabarti, Steven Kordonowy, Niraj Kumar, Sivaprasad Omanakuttan, Michael A Perlin, MS Allman, JP Campora III, Alex Chernoguzov, Samuel F Cooper, et al. Realization of a quantum streaming algorithm on long-lived trapped-ion qubits. *arXiv preprint arXiv:2511.03689*, 2025.
- [Raz99] Ran Raz. Exponential separation of quantum and classical communication complexity. In *STOC*, pages 358–367, 1999.
- [RY15] Anup Rao and Amir Yehudayoff. Simplified lower bounds on the multiparty communication complexity of disjointness. In *CCC*, pages 88–101, 2015.
- [WY26] Haoyu Wang and Guangxu Yang. Exponential quantum space advantage for approximating Max- $k$ SAT in the streaming setting, 2026. arXiv:2606.05366; ECCC TR26-092.
- [YZ24] Guangxu Yang and Jiapeng Zhang. Communication lower bounds for collision problems via density increment arguments. In *STOC*, pages 630–639, 2024.
- [YZ25] Guangxu Yang and Jiapeng Zhang. Deterministic lifting theorems for One-Way Numbers-on-Forehead communication, 2025. arXiv:2506.12420; ECCC TR25-073.
- [YZ26] Guangxu Yang and Jiapeng Zhang. Exponential separation of quantum and classical One-Way Numbers-on-Forehead communication. In *FOCS*, 2026. To appear; arXiv:2603.22795; ECCC TR25-081.